

TASK-1

1.1 TITLE: Conceptual Design using ER Model- Hotel Management System

AIM:

To design the ER diagram for Hotel Management System

Step 1: Problem Understanding and Requirements Analysis

Manage hotel reservations, customer data, room availability, billings and payments efficiently

Requirements:

Customer registration

Room reservation and availability

Billing and invoice generation

Payment tracking

Step 2: Identify major entities

Customer

Hotel

Room

Room category

Payment

Invoice

Bill

Step 3: Identify attributes for each entity

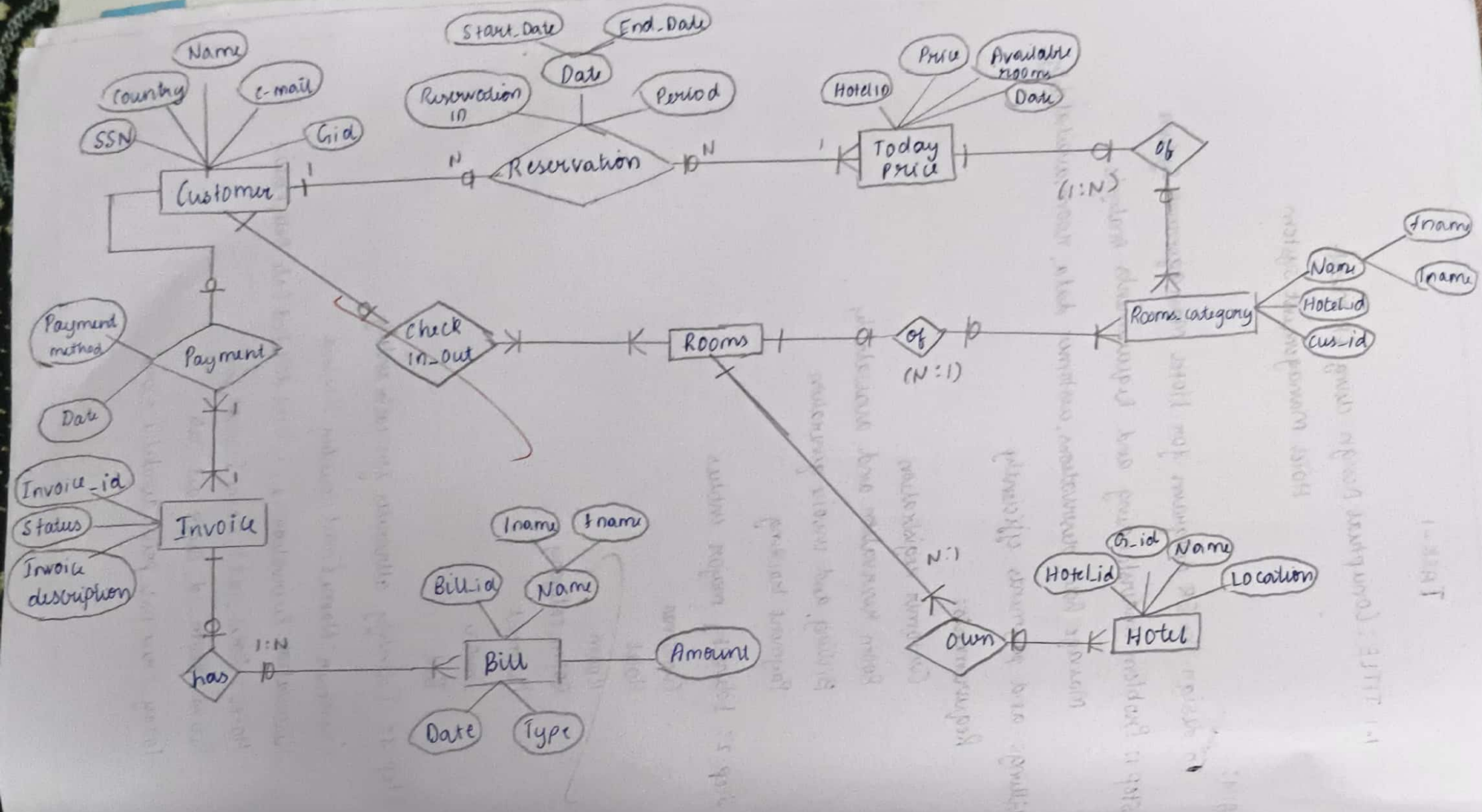
Customer: Name, Email, Country, SSN, Gid

Reservation: Reservation_id(), Start date, End Date, Date, Period

Hotel: Hotel_id(), Name, Location, Gid

Room: Room_id, Check-out, Date

Today: Price, Date, price, available room



Invoice: Invoice_id(1), status, Invoice description

Bill: Bill_id(1), Name, Date, Type, amount

Step 4: Define Relationships between Entities

One customer can have many reservations

One reservation can include many rooms

Many room categories can belong to one hotel

One customer can make many payments

One customer can have multiple invoices

Each invoice is linked to one bill

Many rooms can belong to one room category

Example Relationships

Customer(1) - have $\rightarrow$ (M) Reservations

Reservation(1) - include $\rightarrow$ (M) Rooms

Room categories(M) - belong $\rightarrow$ (1) Hotel

Customer(1) - make $\rightarrow$ (M) Payments

Customer(1) - have $\rightarrow$ (M) invoices

Invoice(1) - linked $\rightarrow$ (1) Bill

Rooms(M) - belong $\rightarrow$ (1) Room category

Renaming the relations with key and constraints

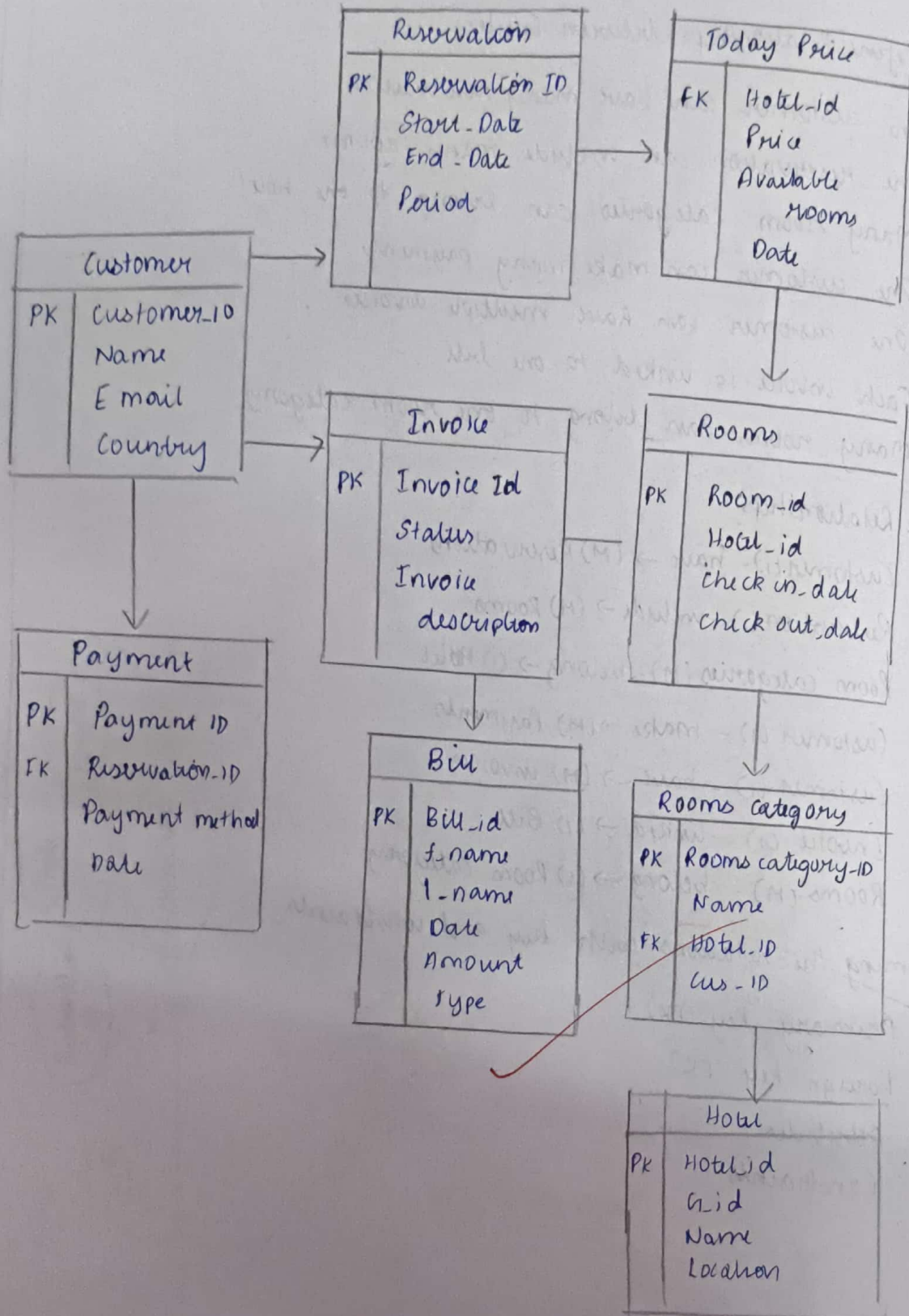
Primary Key (PK)

Foreign key (FK)

Attributes

Constraints

Relational Model



1-2 Convert ER Diagram into Relational Model

AIM:

To design the relational model for Hotel management system

Steps for converting the ER diagram to the table

Entity type becomes a table

All single valued attribute becomes a column for the table

A key attribute of the entity type represented by the primary key

The multi valued attribute is represented by a separate table

Composite attributes represented by components

Derived attributes are not considered in the table

Using these rules we can convert the ER diagram to tables and columns and assign the mapping between the tables

VELTECH	
EX No.	1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VOICE (5)	5
CORD (5)	-
TOTAL (20)	15
INVESTMENT	2017

00817
RESULT:

Thus the ER diagram and conversion of ER diagram to database relational model for Hotel management system has been implemented successfully